Quantum Benchmarks

Hardware Runs

Report issue date: May 4, 2026

This benchmarking report was generated using Classiq v1.10.1.

The report includes hardware runs for a collection of pre-defined and user-defined functions and algorithms, where each benchmark can be instantiated at different problem sizes. For each benchmark experiment, we report a score defined specifically for that example, together with the execution time[1] and quantum-program properties obtained through Classiq's hardware-aware synthesis. For further information, see the Benchmarking directory in classiq-library.

[1]Currently, the reported time includes both execution time and waiting time.

Adder

We benchmark an out-of-place addition by a constant,

$$|x\rangle_n|0\rangle_n \to |x\rangle_n|x + c \bmod 2^n\rangle_n,$$

where n is a the variable size and c is a fixed integer. In the benchmark, the input variable is prepared in a uniform superposition, and the constant is chosen as

$$c = 2^{\lfloor n/2 \rfloor} - 1.$$

The score is defined as the total measured probability of obtaining the correct arithmetic relation,

$$\text{Score} = \sum_{x,y=0}^{2^n-1} P(x,y)\,\delta_{y,\,x+c \bmod 2^n},$$

where $P(x,y)$ is the measured joint output distribution of the two variables. Equivalently, the score is the total probability mass on outcomes satisfying

$$y = x + c \bmod 2^n.$$

A score of 1 corresponds to perfect agreement with the ideal adder transformation.

For more details see Benchmark Adder notebook.

adder - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	9	102	84
Classiq	simulator_statevector	1.0	0.0	9	103	84

GHZ State Preparation

We benchmark the preparation of the n-qubit GHZ state,

$$\frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}.$$

Following the population-and-parity fidelity estimator of Monz et al., PRL 106, 130506 (2011), the score combines the population $P = p_0 + p_{2^n-1}$ with the coherence C, extracted from the amplitude of the parity oscillation $\Pi(\phi) = P_{\text{even}}(\phi) - P_{\text{odd}}(\phi)$ under a collective analysis rotation. The resulting fidelity estimator is

$$F = \frac{P + C}{2}.$$

For more details see Benchmark GHZ notebook.

ghz - 3 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	3	4	2
Classiq	simulator_statevector	1.0	0.0	3	5	2

Grover Search Algorithm

We benchmark Grover's search algorithm on n qubits with a single marked state $|C\rangle$. The algorithm applies

$$r = \left\lfloor \frac{\pi}{4\theta} - \frac{1}{2} \right\rfloor$$

Grover iterations, where

$$\theta = \arcsin\left(\frac{1}{\sqrt{2^n}}\right),$$

yielding the ideal success probability

$$P_{\text{ideal}} = \sin^2\big((2r+1)\theta\big).$$

The score is defined as the ratio of the measured success probability to the ideal one,

$$S = \frac{P_{\text{good}}}{P_{\text{ideal}}}.$$

This benchmark also probes the correct implementation of a multi-controlled X operation, which appears in the Grover oracle and diffusion step.

For more details see Benchmark Grover notebook.

grover - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	0.9896	0.0	5	168	82
Classiq	simulator_statevector	1.0	0.0	6	114	58

Dynamical Localization

We benchmark dynamical localization in the quantum sawtooth map on 3 qubits. Because this genuinely quantum effect relies on the coherent accumulation of phase over several time steps, while still admitting a relatively simple circuit implementation, it has been proposed as a clear benchmarking task for quantum hardware in Pizzamiglio et al., Entropy 23, 654 (2021). The system is initialized in the momentum eigenstate $|\psi_0\rangle = |-2\rangle$ and evolved for T kicks. Let $P_T(-2)$ denote the measured probability of the initial momentum after T kicks, and let $N = 2^3 = 8$. The normalized peak strength is

$$\tilde{P}_T = \frac{P_T(-2) - 1/N}{1 - 1/N},$$

clipped to $[0, 1]$. The score is defined as the geometric mean over two time steps,

$$S = \sqrt{\tilde{P}_2 \tilde{P}_3}.$$

For more details see Benchmark Dynamical Localization notebook.

localization - 3 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	0.9297	0.0	3	59	40
Classiq	simulator_statevector	0.9281	0.0	3	62	40

QFT

The Quantum Fourier Transform (QFT) function is a quantum algorithmic primitive, utilized in algorithms such as Shor's Factoring, Discrete Logarithm and Quantum Phase Estimation. It is the quantum analog of the discrete Fourier transform, applied to a quantum state vector. We benchmark the QFT on n qubits by preparing the equal superposition (comb) state

$$|\psi_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{t=0}^{2^n-1} |t\rangle = H^{\otimes n}|0^n\rangle,$$

applying the QFT, and comparing the measured output distribution to the expected outcome $|\psi_f\rangle = |0^n\rangle$. The score is defined from the total variation distance between the observed output distribution p and the ideal distribution u,

$$D_{\text{TV}}(p,u) = \frac{1}{2} \sum_{i=0}^{2^n-1} |p_i - u_i|.$$

We then define the score:

$$S = 1 - \frac{D_{\text{TV}}(p,u)}{1 - 1/n}.$$

(This example can be run with different comb density of size m: $|\psi_0\rangle = \frac{1}{\sqrt{2^{N-m}}} \sum_{t=0}^{2^{N-m}-1} |t2^m\rangle$.)

For more details see Benchmark QFT notebook.

qft - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	4	23	18
Classiq	simulator_statevector	1.0	0.0	4	24	18

Linear State Preparation

We benchmark the preparation of an n-qubit state with linearly increasing amplitudes,

$$|\psi\rangle = \frac{1}{\sqrt{M}} \sum_{x=0}^{2^n-1} x\,|x\rangle,$$

where

$$M = \sum_{x=0}^{2^n-1} x^2.$$

The score is defined from the total variation distance between the observed output distribution p and the ideal distribution u,

$$D_{\mathrm{TV}}(p,u) = \frac{1}{2} \sum_{i=0}^{2^n-1} |p_i - u_i|.$$

We then define

$$S = 1 - \frac{D_{\mathrm{TV}}(p,u)}{1 - 1/n}.$$

For more details see Benchmark State Preparation notebook.

state_preparation - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	0.9544	0.0	4	50	28
Classiq	simulator_statevector	1.0	0.0	4	51	28

Quantum Volume

We benchmark Quantum Volume following Cross et al., PRA 100, 032328 (2019). For each width n, we generate square model circuits of width n and depth n, and measure the heavy-output probability h, namely the total probability assigned to outcomes whose ideal probabilities are above the median of the ideal output distribution.

A width is considered to pass if the heavy-output probability exceeds the threshold 2/3 according to a statistical lower bound. For runs with at least 100 completed trials, we use the pooled confidence criterion of Cross et al. If n_c is the number of completed circuits, n_s the number of shots per circuit, n_h the total number of heavy outputs observed across all trials, and σ the chosen confidence parameter, we use

$$h_{\mathrm{LB}} = \frac{n_h - \sigma\sqrt{n_h(n_c n_s - n_h/n_c)}}{n_c n_s},$$

and declare the width successful when $h_{\mathrm{LB}} > 2/3$. For smaller runs, we use the simpler normal approximation

$$h_{\mathrm{LB}} = \bar{h} - \sigma\,\mathrm{SE}(\bar{h}),$$

where $\bar{h}$ is the sample mean heavy-output probability across trials and $\mathrm{SE}(\bar{h})$ is its standard error.

The reported Quantum Volume is

$$QV = 2^{n_{\max}},$$

where $n_{\max}$ is the largest tested width that passes. Consequently, if the largest executed width also passes, the true quantum volume is at least this value and may in fact be larger.

For more details see Quantum Volume notebook.

2–4 qubits (10 trials, 1000 shots)

Provider	Backend	Quantum Volume	Trials	Average Elapsed Time (min)
Classiq	simulator	16	30/30	0.01
Classiq	simulator_statevector	16	30/30	0.01